research note 3

1 Mathematica Source Code

```
ClearAll["Global`*"];
Needs["FeynCalc`"];

(*=========================================*)
(*1. define propagator & rules*)
(*=========================================*)
denS[s_] := s - MA^2;
denT[t_] := t - MA^2;
denU[u_] := u - MA^2;

GetMandelstamRules[s_, t_, u_, m1_, m2_, m3_, m4_] :=
  FCI[{SP[p1, p1] -> m1^2, SP[p2, p2] -> m2^2, SP[p3, p3] -> m3^2,
    SP[p4, p4] -> m4^2, SP[p1, p2] -> (s - m1^2 - m2^2)/2,
    SP[p2, p1] -> (s - m1^2 - m2^2)/2,
    SP[p3, p4] -> (s - m3^2 - m4^2)/2,
    SP[p4, p3] -> (s - m3^2 - m4^2)/2,
    SP[p1, p3] -> (m1^2 + m3^2 - t)/2,
    SP[p3, p1] -> (m1^2 + m3^2 - t)/2,
    SP[p2, p4] -> (m2^2 + m4^2 - t)/2,
    SP[p4, p2] -> (m2^2 + m4^2 - t)/2,
    SP[p1, p4] -> (m1^2 + m4^2 - u)/2,
    SP[p4, p1] -> (m1^2 + m4^2 - u)/2,
    SP[p2, p3] -> (m2^2 + m3^2 - u)/2,
    SP[p3, p2] -> (m2^2 + m3^2 - u)/2}];

ProcessCleanUp[rawMatrix_, rule_, elimRule_] :=
  rawMatrix // ChangeDimension[#, 4] & // ExpandScalarProduct //
      ReplaceAll[#, rule] & // ReplaceAll[#, elimRule] & // Simplify;

(*=========================================*)
(*2. matrix elements (Process 1& 2)*)
(*=========================================*)
rule1 = GetMandelstamRules[s, t, u, mchi, mchi, mchi, mchi];

(*Process 1*)
M1s = gchi^2/denS[s]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorV[p4, mchi];
M1t = gchi^2/denT[t]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]*
   SpinorVBar[p2, mchi] . GA[\[Mu]] . SpinorV[p4, mchi];
M1 = M1s - M1t;
M1sqRaw =
  FermionSpinSum[M1*ComplexConjugate[M1]]/4 // Contract //
```

```
    DiracSimplify;
M1sq = ProcessCleanUp[M1sqRaw, rule1, u -> 4 mchi^2 - s - t];

(*Process 2*)
M2t = gchi^2/denT[t]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]*
   SpinorUBar[p4, mchi] . GA[\[Mu]] . SpinorU[p2, mchi];
M2u = gchi^2/denU[u]*
   SpinorUBar[p4, mchi] . GA[\[Mu]] . SpinorU[p1, mchi]*
   SpinorUBar[p3, mchi] . GA[\[Mu]] . SpinorU[p2, mchi];
M2 = M2t - M2u;
M2sqRaw =
  FermionSpinSum[M2*ComplexConjugate[M2]]/4 // Contract //
    DiracSimplify;
M2sq = ProcessCleanUp[M2sqRaw, rule1, s -> 4 mchi^2 - t - u];

(*sigmaT integral*)

kinematicRules = {t -> -(s - 4 mchi^2)/2*(1 - z),
   u -> -(s - 4 mchi^2)/2*(1 + z)};
dSigmaDz[Msq_] := (Msq /. kinematicRules)/(32 Pi s);

SigmaT1 =
  Integrate[dSigmaDz[M1sq]*(1 - z), {z, -1, 1},
    Assumptions -> {s > 4 mchi^2, MA > 0, mchi > 0}] // Simplify;
Print["\[Sigma]_T (Process 1 integ finished) = ", SigmaT1];

SigmaT2 =
  1/2*Integrate[dSigmaDz[M2sq]*(1 - z), {z, -1, 1},
     Assumptions -> {s > 4 mchi^2, MA > 0, mchi > 0}] // Simplify;
Print["\[Sigma]_T (Process 2 integ finished) = ", SigmaT2];

NRRule = {s -> 4 mchi^2 + mchi^2 v^2};
AlphaRule = {gchi^4 -> 16 Pi^2 alphaX^2};

SigmaT1NR = SigmaT1 /. NRRule /. AlphaRule;
SigmaT2NR = SigmaT2 /. NRRule /. AlphaRule;

(*introduce R:by MA->(mchi*v)/R*)
RRule = {MA -> (mchi*v)/R};
SigmaT1WithR = SigmaT1NR /. RRule;
SigmaT2WithR = SigmaT2NR /. RRule;


SigmaT1Base =
  Series[SigmaT1WithR, {v, 0, 0}] // Normal // FullSimplify;
SigmaT2Base =
  Series[SigmaT2WithR, {v, 0, 0}] // Normal // FullSimplify;

Print["========================================="];
Print["\[Sigma]_T (Process 1) mother formyla with R = "];
Print[SigmaT1Base];
Print["\[Sigma]_T (Process 2) mother formula with R = "];
Print[SigmaT2Base];
Print["========================================="];

```

```
(*heavy mediator limit (R <<1)*)
SigmaT1Heavy =
  Series[SigmaT1Base, {R, 0, 4}] // Normal // FullSimplify;
SigmaT2Heavy =
  Series[SigmaT2Base, {R, 0, 4}] // Normal // FullSimplify;

Print["\[Sigma]_T (Process 1, heavy mediator R << 1) = "];
Print[SigmaT1Heavy /. R -> (mchi*v)/MA];
Print["\[Sigma]_T (Process 2, heavy mediator R << 1) = "];
Print[SigmaT2Heavy /. R -> (mchi*v)/MA];
Print["========================================="];

(*light mediator limit (R>>1)*)
SigmaT1Light =
  Series[SigmaT1Base, {R, Infinity, 0}] // Normal // FullSimplify;
SigmaT2Light =
  Series[SigmaT2Base, {R, Infinity, 0}] // Normal // FullSimplify;

Print["\[Sigma]_T (Process 1, light mediator R >> 1) = "];
Print[SigmaT1Light /. R -> (mchi*v)/MA];
Print["\[Sigma]_T (Process 2, light mediator R >> 1) = "];
Print[SigmaT2Light /. R -> (mchi*v)/MA];
Print["========================================="];
Print["---> All calculations finished "];
```

2 Results

Initial states average, final states sum. My processes 6,7 are neutrinos to dark matters. In 2603.03444 it is dark matters to neutrinos. So in that case should divide by 4. But here, the initial state is left hand helicity state. State is all internal DoF that are not measured or controlled by the experiment.

Here $|\nu_i\rangle = -\sin\frac{\theta}{2}|\uparrow\rangle_z + e^{i\varphi}\cos\frac{\theta}{2}|\downarrow\rangle_z$***,*** $\hat{p}$ ***is determined, so*** (θ,φ)***is determined,*** $|\nu_i\rangle$ ***is unique. Although there are spin basis*** $|\uparrow\rangle_z and|\downarrow\rangle_z$***, the physical freedom is 1.***

Hence, when averaging initial states, should divide by the number of total states for the 2 particles, which is 1.

Also,I didn't use the viscosity cross section $\int\frac{d}{d}d\sin^2$ ***mentioned in paper 1709.07882. For identical particles, I used*** $\frac{1}{2T}$***. If use viscosity cross section, answers may be different.***

2.1 General Non-Relativistic Mother Formula

The general scattering cross section for Process 1 & 2 parameterized with the dimensionless ratio $R = m_\chi v/M_A$ is given by:

$$\begin{aligned}\sigma_T^{\text{Process 1}} =& -\frac{\pi\alpha_X^2}{4m_\chi^2R^4(R^2+1)v^4}\Bigg[R^6(-7v^4+80v^2+32)+R^4v^2(21v^2+88)+34R^2v^4\\&+2(R^2+1)\left(R^4(v^4+12v^2+16)+2R^2v^2(v^2+22)+17v^4\right)\ln\left(\frac{1}{R^2+1}\right)\Bigg]\\ \sigma_T^{\text{Process 2}} =&\frac{\pi\alpha_X^2}{4m_\chi^2R^2(R^4+3R^2+2)v^4}\Bigg[R^2(R^2+2)(R^4(v^4+12v^2+16)+2R^2v^2(v^2+8)+4v^4)\\&-(R^2+1)(R^4(3v^4-12v^2-16)-4R^2v^2(v^2+8)-8v^4)\ln\left(\frac{1}{R^2+1}\right)\Bigg]\end{aligned} \tag{1}$$

2.2 Heavy Mediator Limit ($R \ll 1$)

In the heavy mediator limit ($M_A \gg m_\chi v,\ m_\chi v \ll M_A \ll m_\chi$), the expanded cross sections yield:

Process 1 ($\chi\bar{\chi}\to\chi\bar{\chi}$):

$$\sigma_T^{R\ll1} = \frac{\pi\alpha_X^2(4v^2-9)}{3M_A^2} + \frac{\pi\alpha_X^2m_\chi^2(96+56v^2-13v^4)}{24M_A^4} \tag{2}$$

Process 2 ($\chi\chi\to\chi\chi$):

$$\sigma_T^{R\ll1} = \frac{\pi\alpha_X^2m_\chi^2(19v^4+36v^2+48)}{24M_A^4} \tag{3}$$

Because $v \to 0$,

$$\begin{aligned}\sigma_{T1}^{R\ll 1} &= \frac{\pi\alpha_X^2 M_A^2(-72)}{24M_A^4} + \frac{\pi\alpha_X^2 m_\chi^2 96}{24M_A^4} = \frac{\pi\alpha_\chi^2}{M_A^4}(4m_\chi^2 - 3M_A^2) = \frac{4m_\chi^2\pi\alpha_\chi^2}{M_A^4} \\ \sigma_{T2}^{R\ll 1} &= \frac{2m_\chi^2\pi\alpha_\chi^2}{M_A^4}\end{aligned} \tag{4}$$

2.3 Light Mediator Limit ($R \gg 1$)

In the light mediator limit ($M_A \ll m_\chi v$), the logarithmic Coulomb enhancement emerges:

Process 1 ($\chi\bar{\chi} \to \chi\bar{\chi}$):

$$\sigma_T^{R\gg 1} = \frac{\pi\alpha_X^2}{4m_\chi^2 v^4}\left[4(v^4 + 12v^2 + 16)\ln\left(\frac{m_\chi v}{M_A}\right) + 7v^4 - 80v^2 - 32\right] \tag{5}$$

Process 2 ($\chi\chi \to \chi\chi$):

$$\sigma_T^{R\gg 1} = \frac{\pi\alpha_X^2}{4m_\chi^2 v^4}\left[\frac{m_\chi^2 v^2(v^4 + 12v^2 + 16)}{M_A^2} + (6v^4 - 24v^2 - 32)\ln\left(\frac{m_\chi v}{M_A}\right) + v^4 + 4v^2 - 16\right] \tag{6}$$

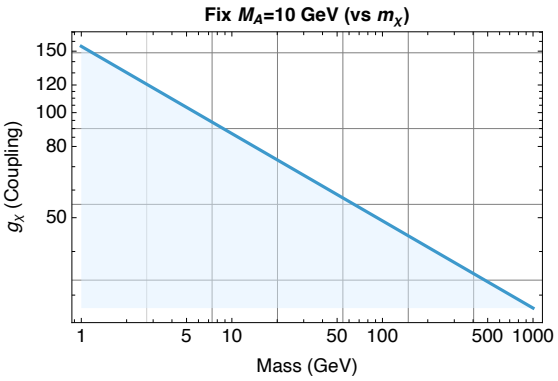
Because $v \to 0$,

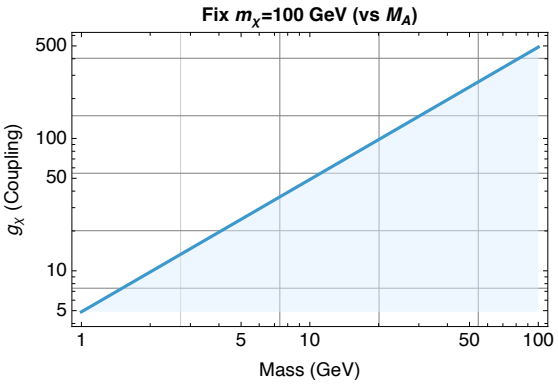
$$\begin{aligned}\sigma_{T1}^{R\gg 1} &= \frac{16\pi\alpha_\chi^2}{m_\chi^2 v^4}\ln(\frac{m_\chi v}{M_A}) \\ \sigma_{T2}^{R\gg 1} &= \frac{\pi\alpha_\chi^2}{m_\chi^2 v^4}[4R^2 - 8\ln R] = \frac{\pi\alpha_\chi^2}{m_\chi^2 v^4}[4(\frac{m_\chi v}{M_A})^2 - 8\ln(\frac{m_\chi v}{M_A})] =? = \frac{4\pi\alpha_\chi^2}{M_A^2 v^2}\end{aligned} \tag{7}$$

Here I am not sure if I should ignore log term. It is small compare with R^2, but I think it may reflect some special behaviors.

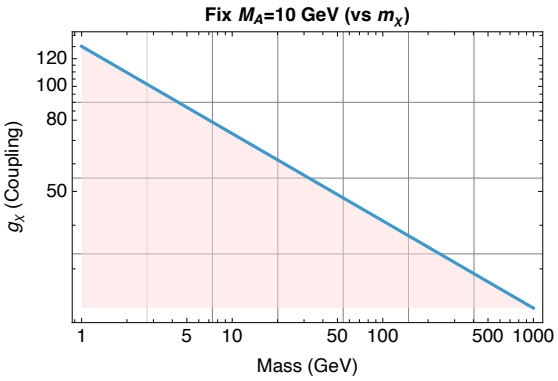
Another thing I worry about is some vs could be expressed in R, or some vs inside R could be moved out and makes the final limit expression different. Now I followed substitute $M_A \to \frac{m_\chi v}{R}$ and then expand at $v = 0$ to 0th order and then expand at R=0 or ∞ at 4th order.

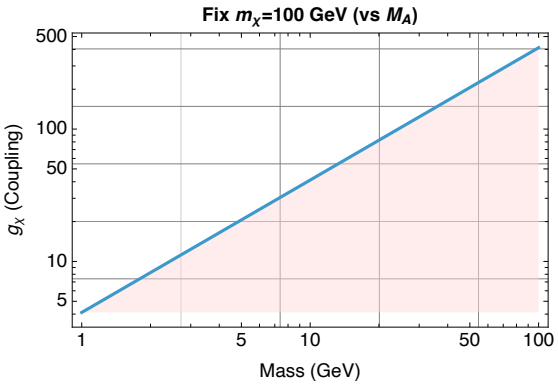
Process 1 (Heavy Mediator)





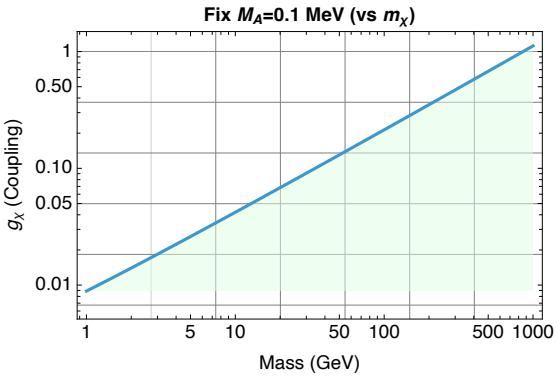
Process 2 (Heavy Mediator)

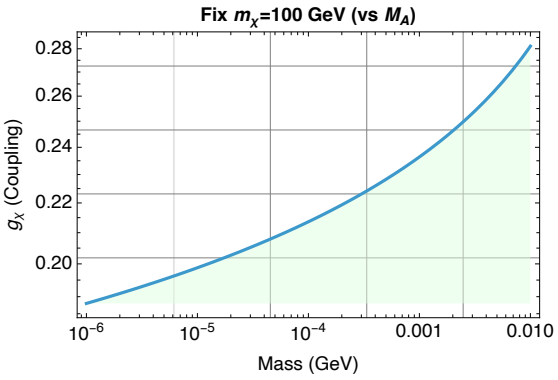




--- Light Mediator ---

Process 1 (Light Mediator)





Process 2 (Light Mediator)

